

USN

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester MCA (CBCS) Degree Examinations
 December 2024

22MCA343 – BLOCKCHAIN TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Define Blockchain. Explain the working of Blockchain with a neat diagram.	10	L*2	1	1
b)	List and explain the myths about Bitcoin.	10	L2	1	1
2. a)	Explain the major pros and cons of Blockchain Technology.	10	L2	1	1
b)	Explain with an example the working of Public and Private keys in Blockchain Technology.	10	L2	1	1
Unit – II					
3. a)	Explain the core components of Blockchain Architecture with a neat diagram.	8	L2	2	1
b)	Explain the different variants of Blockchain Architecture.	12	L2	2	1
4. a)	What is Cryptocurrency? Explain different types of Cryptocurrency wallets.	10	L2	2	1
b)	What are the key differences between traditional databases and blockchain technology? And how does Bitcoin differ from blockchain as a technology?	10	L2	2	1
Unit – III					
5. a)	Explain Double spending in Blockchain and how bitcoin handles double spending.	10	L2	3	1
b)	Explain the concept of Merkle Tree.	10	L2	3	1
6. a)	What is hashing in Blockchain. Explain the characteristics of Cryptographic hash function.	10	L2	3	1
b)	Explain the different types of Blockchain mining and how the conflicts are resolved in bitcoin network.	10	L2	3	1
Unit – IV					
7. a)	Define Bitcoin. Explain with example the general steps of sending and receiving the Bitcoins.	10	L2	4	1
b)	Explain the advantages and disadvantages of using bitcoin.	10	L2	4	1
8. a)	Explain how to choose a Bitcoin wallet.	10	L2	4	1
b)	Explain the different ways to convert Bitcoin to Fiat currency.	10	L2	4	1
Unit – V					
9. a)	What is Ethereum? Explain the benefits of Ethereum?	8	L2	5	1
b)	Explain with a neat diagram the working of smart contracts with its advantages.	12	L2	5	1
10. a)	Compare Bitcoin and Ethereum.	10	L2	5	1
b)	Define Crypto Contract. Explain the Features of Crypto contract.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
